

Developing a Solution – SGBA+

Group 1

MEDSCIEN9503

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August 07, 2026

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Assignment Acknowledgement**Author Contributions**

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To address the pervasive gaps in medical research, we propose a comprehensive two-part solution designed to transition to an evidence-based model of individualized medicine that recognizes biological variations as a fundamental component of drug development.

Part I: Strengthening Mandates and Evidence-Based Recalibration

The first pillar of this solution involves the implementation of stricter mandates for Sex as a Biological Variable (SABV) across drug development. Current policies allow for inconsistent enforcement and a “minimal compliance” culture where researchers include both sexes to obtain funding but fail to conduct meaningful disaggregated analysis (Farrukh & Becker, 2025).

Current human research policies across the globe have encouraged the inclusion of women and minority populations; however, these guidelines rarely apply to the preliminary studies conducted on animals, which ultimately influence how human trials are conducted. The US National Institute of Health (NIH) first encouraged the inclusion of women in research in 1989, yet inclusion did not become a legal requirement for NIH-funded clinical research until the NIH Revitalization Act of 1993 (Ahmed et al., 2026). Similarly, the European Union introduced the Gender Watch system through its Fifth Framework Programme in 1998 to improve the collection of sex-disaggregated data and strengthen gender-sensitive research practices (*Report on Women and Science, 2001*). Despite these policy advancements, there remain two key issues.

1. Inadequate enforcement and accountability have limited their effectiveness, demonstrating that inclusion alone is insufficient without mechanisms ensuring rigorous sex-specific analysis throughout the research process.
2. Policies rarely apply to the preliminary animal studies that influence the human trials, propagating the male-centered norm from historical unisex data.

Part II: Revision of Regulations Using Updated Evidence

The second pillar focuses on translating the evidence generated through sex inclusive preclinical research into lasting changes within Canada’s regulatory framework. Part II ensures that the newly collected evidence actively informs how new therapeutics are evaluated, approved, and ultimately prescribed. By redefining the evidentiary standard from one based on an average patient to one that explicitly accounts for biological variation and individualization, Health Canada can shift drug development away from a one-size-fits-all model.

The proposed solution is implemented through two complementary policy levers that target different stages of drug development. The first phase uses federal research funding as the primary policy lever. By strengthening CIHR grant requirements, researchers receiving public funding would be required to include both male and female animal models in preclinical studies, conduct sex-disaggregated analyses, and report these findings as a condition of continued funding. The second phase focuses on embedding sex-inclusive evidence into Canada’s regulatory framework to redefine the biomedical reference standard. Once sufficient evidence has been generated through Part I, Health Canada can establish a regulation that views biological variability as a core component of rewriting historical norms.

Challenges to be Addressed

Part I addresses the minimal-compliance culture, a result of current mandates. Rotz and colleagues (2022) interviewed researchers who implied that simply checking off male and female criteria for their population was enough to have considered gender and equity issues.

Even where women are included, biological sex is often treated only as a demographic characteristic rather than as a dynamic variable, so clinically meaningful sex-specific differences remain undetected (Miller, 2014; Sarri & Hoiseth, 2026). Chen et al. (2026) found that reviewer-level SGBA requirements more effectively shifted funding behaviour than application-level requirements. Research ethics committees compound this, lacking tools to assess how sex, gender and intersectional dimensions should be integrated, so brief mentions of sex in a protocol are often accepted as sufficient (Delgado et al., 2025). Part I's monitoring and auditing requirement converts a declaration made in an application into a verifiable obligation.

A second barrier is that trials are rarely designed to answer sex-specific questions. Most are not powered to detect sex-specific outcomes, so findings are generated post hoc through underpowered subgroup analyses (Carland et al., 2022). Exclusion criteria built around anticipatory motherhood further narrow eligibility even for participants who are not pregnant and do not intend to become pregnant (Cottingham & Fisher, 2022). By intervening at the funding and design stage, Part II addresses the translation gap. Sex differences in disease presentation are well documented, yet their translation into sex-specific diagnostic frameworks and treatment remains limited, and the scarcity of sex- and gender-disaggregated data makes it difficult to address disparities in diagnostic accuracy or develop evidence-based solutions (Upadhyay et al., 2025). Mandatory guideline review closes the loop between new evidence and clinical standards, aligning with Health Canada's (2022) use of disaggregated data to verify that trial participants reflect those who will actually use a product.

Critically, these barriers are neither disease-specific nor resolved by enrolling more women alone. Female representation falls short of disease burden across oncology, neurology, immunology and nephrology (Steinberg et al., 2021), and women of Asian, Black and Hispanic descent were less likely than white women to enroll in gynecological cancer trials, with white women overrepresented (Khadraoui et al., 2023). A solution operating through funding conditions and guideline review therefore applies across therapeutic areas and accommodates the intersectional dimension of SGBA+.

Accordance with Health Canada's Mandate

Health Canada's mandate (2004) illustrates its role in "protect[ing] Canadians from unsafe food, health and consumer products". More specifically, they are federal regulators for many health care products, including drugs, medical devices, and natural health products (Health Canada, 2017). Regulation occurs in a few different domains. First is assessing the safety, efficacy, and quality of drugs (Health Canada, 2002), typically through monitoring drugs in the current market. They analyze safety reviews and adverse reaction reports to better understand the effects of these therapeutics from sample to population (Health Canada, 2017). Secondly, they aid in regulating the clinical trial and drug approval process. Health Canada requires pharmaceutical companies to submit their results of clinical testing, samples that were tested, and their application for product approval. With this information, they assess whether the benefits outweigh the risks or if the evidence supports the health claims made by these companies. Ultimately, Health Canada makes the decision: if they should recommend, request a change to, or remove the therapeutic from the market. Beyond drug approval, they conduct inspections during clinical trials to ensure compliance with the standards and regulations of Health Canada (Health Canada, 2017). Lastly, regulation applies to future guidance towards drug development research. They have identified a need to increase transparency with clinical trials to the public to

improve recruitment of underrepresented populations, such as women, indigenous populations, ethnic, and racialized groups (Health Canada, 2026).

With the risk of adverse drug reactions, decreased treatment efficacy in women, and the need for diagnostic criteria that consider sex, all studies should follow the SABV mandate and incorporate data acknowledging sex differences. Yet, gaps in this framework allow researchers to overlook data analysis by sex, sex-specific effects, and whether sex plays a role in modifying treatment outcomes (Farrukh & Becker, 2025). If Health Canada supports stricter mandates for this policy through Part I of our solution under its regulatory role in drug development, more meaningful investigations can be done. Sex-specific effects that would have otherwise gone undetected can be reported, which may affect the quality, applicability, and safety of drugs being supplied to Canadians. Additionally, only through this can further variables like race, age, socioeconomic status, and disability be measured through Part II. Thus, Health Canada can ensure drugs authorized in Canada are supported by evidence that adequately shows benefits for the populations it is intended to serve.

Key Stakeholders and Anticipated Reactions

The key stakeholders for this project include researchers, funding agencies such as the Canadian Institutes of Health (CIHR), clinicians, and patient advocacy groups. Researchers are the most impacted group because the proposed policy changes affect how preclinical research is conducted. Initially, based on past reactions, researchers may view these requirements as onerous and burdensome, potentially due to increased costs, resource demands, and methodological complexity (Miller et al., 2016; Tannenbaum et al., 2016). However, enforcing stricter mandates for SABV and requiring researchers to move beyond the simple inclusion of female animals and conduct meaningful sex-disaggregated analyses will improve scientific rigour and translational value of drug development.

Funding agencies, such as CIHR, play a central role, helping implement these mandates through grant approvals and ensuring researchers meaningfully incorporate SABV. CIHR's anticipated reaction should be supportive because they have recognized SABV as an essential component in producing clinically relevant research since 2010 (Government of Canada, Canadian Institutes of Health Research, 2016). It may, however, require additional oversight and administrative resources from their end. Similarly, regulatory agencies such as Health Canada and the Food and Drug Administration (FDA) are expected to respond positively as stronger preclinical research improves evaluation of drug safety and efficacy before reaching clinical trials. As drug development moves away from the male-centric norm, the improved translation of preclinical findings will provide clinicians with robust evidence to guide patient care (Arnegard et al., 2020). Consequently, healthcare providers would likely support this project because stronger sex-inclusive research can better inform diagnosis, prescribing, and treatment plans (Arnegard et al., 2020).

Lastly, patient advocacy groups, such as the Society for Women's Health Research (SWHR), are anticipated to be strong supporters of the proposal. For organizations such as SWHR, this proposal aligns with their position that research should move beyond simply including both sexes to conducting adequately powered sex-based analyses (Society for Women's Health Research, 2020). Overall, patient advocacy groups would view the proposal favourably, because it addresses longstanding inequities in drug development.

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